SAIGON AI HUB OneVoice AI Challenge Qualcomm

OneVoice AI Challenge Deck
DiffusPeak Team

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1 Introduction

In today's highly globalized economy, language barriers remain a significant obstacle to productivity, safety, and career mobility, particularly in industrial environments such as factories, construction sites, and logistics hubs. Statistics show that a vast majority of workers in developing countries like Vietnam lack proficiency in foreign languages, yet they frequently collaborate with international experts and managers. Traditional cloud-based translation solutions often fail in these environments due to unstable internet connectivity, high latency, and privacy concerns.

To address this critical gap, the **OneVoice AI Challenge** calls for the development of next-generation, on-device translation systems powered by Edge AI. Our objective in this challenge is to design a highly efficient Speech Translation model that facilitates seamless, real-time bilingual communication. The system must operate entirely on edge devices without cloud dependency, targeting three primary language pairs: Vietnamese ↔ English, Vietnamese ↔ Mandarin, and Vietnamese ↔ Korean. The core challenge lies in balancing the trade-off between translation accuracy and the strict hardware constraints of edge devices, specifically requiring ultra-low latency and minimal memory footprint. Our implementation is open-source and publicly available.[1]

2 Problem Formulation

The task of Edge-based Speech Translation can be mathematically formulated as a constrained sequence-to-sequence generation problem.

Let $\mathbf{X} = (x_1, x_2, \ldots, x_{T_x})$ represent the sequence of acoustic features (e.g., log-Mel spectrograms or raw audio representations from a pre-trained encoder like Moonshine or Wav2Vec2) extracted from the source speech. Let $\mathbf{Y} = (y_1, y_2, \ldots, y_{T_y})$ be the corresponding sequence of translated text tokens in the target language.

[1] https://github.com/mainhatduy/diffusion-speech-recognition

For a standard autoregressive end-to-end speech translation model parameterized by θ, the objective is to model the conditional probability of the target sequence given the source audio:

$$P(\mathbf{Y}|\mathbf{X};\theta) = \prod_{i=1}^{T_y} P(y_i|y_{<i}, \mathbf{X};\theta) \tag{1}$$

The optimal model parameters θ^* are obtained by minimizing the negative log-likelihood (NLL) over the training dataset $\mathcal{D}$:

$$\mathcal{L}_{\text{ST}}(\theta) = \mathbb{E}_{(\mathbf{X},\mathbf{Y})\sim\mathcal{D}} \left[-\sum_{i=1}^{T_y} \log P(y_i|y_{<i}, \mathbf{X};\theta) \right] \tag{2}$$

3 Methodology

The proposed model leverages absorbing discrete diffusion with a pre-trained Masked Language Model (RoBERTa) as the core decoder. For multimodal fusion, it employs a Moonshine acoustic encoder to extract continuous audio features. These features are linearly projected and concatenated as a prefix to the text embeddings. This prefix conditioning enables the Transformer to compute global self-attention across both modalities seamlessly.

Unlike continuous diffusion, our forward process corrupts the ground-truth sequence $x_0 = \{x_0^1, \dots, x_0^N\}$ in a discrete space. Specifically, each token is independently replaced by a `[MASK]` token ($[M]$) with a probability linearly correlated with the timestep $t \in [0, T]$:

$$q(x_t^i = [M]|x_0^i) = \frac{t}{T}, \quad q(x_t^i = x_0^i|x_0^i) = 1 - \frac{t}{T} \tag{3}$$

Conversely, the reverse denoising process operates non-autoregressively. It initializes with a fully masked sequence $x_T = \{[M], \dots, [M]\}$ and iteratively recovers the text. At each step t, the model predicts the original tokens $p_\theta(x_0|x_t, \text{audio})$ and updates x_{t-1} by unmasking the most confident tokens. This process follows a dynamic cosine schedule and utilizes reparameterized decoding to efficiently converge to the final prediction.

3.1 Highlights Compared to Existing Models

The proposed framework offers several distinct advantages over traditional speech recognition paradigms:

- **Parallelized Generation:** Fundamentally different from autoregressive (AR) models like Whisper, the diffusion model predicts and updates multiple tokens simultaneously. This non-autoregressive (NAR) characteristic significantly reduces inference latency for extended audio sequences.

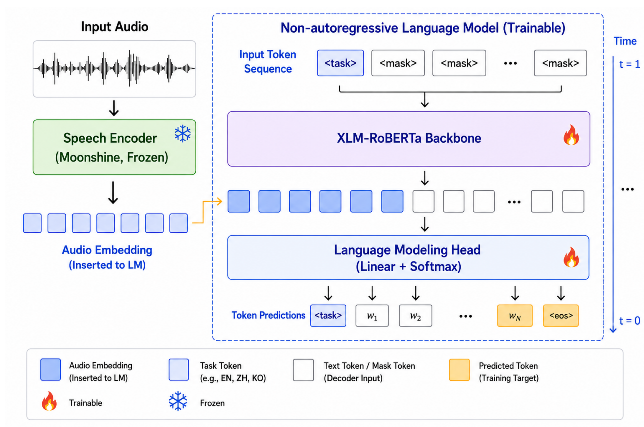

Figure 1: Architecture

- **Pre-trained MLM Alignment:** The absorbing diffusion objective naturally aligns with MLM pre-training. Consequently, the model directly leverages RoBERTa's robust bidirectional language modeling capabilities without retraining an AR generation structure from scratch.

- **Unmasked Information Flow:** By eliminating causal masking, each token globally attends to all acoustic features and surrounding text elements at every step. This bidirectional cross-modal attention optimizes the alignment between audio and text, further enhancing recognition accuracy.

- **Dynamic Iteration Steps:** The number of generation steps can be flexibly adjusted during inference. Executing more steps yields higher recognition accuracy, while using fewer steps significantly accelerates inference speed, allowing for a controllable trade-off between performance and latency.

- **Single-Device Inference Optimization:** The model architecture is optimized for efficient inference entirely on a single device. Unlike autoregressive (AR) models that suffer from underutilized compute capacity due to sequential token generation, our non-autoregressive approach processes tokens in parallel, thereby maximizing hardware resource utilization. This minimizes hardware constraints and streamlines deployment without relying on complex, multi-model distributed pipelines.

- **Self-Correction Capability:** Driven by the iterative denoising process, the diffusion model possesses an intrinsic self-correction mechanism. Errors or uncertain predictions from early timesteps are continuously evaluated and rectified in subsequent iterations, leading to robust and refined outputs.

4 Experiments

4.1 Dataset and Experiment Setup

To train and evaluate the proposed system, we utilize the **VietSpeech** dataset [4]. Because the original corpus primarily focuses on Vietnamese speech recognition and lacks multilingual text targets, we employ a high-quality machine translation model, specifically **Hy-MT2-30B** [5], to synthesize the ground-truth translation labels. The resulting multilingual training dataset, consisting of translations into English (EN), Mandarin (ZH/CN), and Korean (KO), has been released as **VietSpeech-Train-Translated** [1].

The backbone of our framework is built upon the **Discrete Diffusion Language Model** architecture, which facilitates highly parallelized non-autoregressive decoding. For acoustic feature extraction, we integrate the **Moonshine** (*streaming-medium*) audio encoder [3]. This architecture combination ensures robust cross-modal representation learning while maintaining strict computational efficiency suitable for edge environments.

4.2 Experiment Results

The model is evaluated on the **VietSpeech-Validation-Translated** dataset [2], comprising 10,261 audio samples with a total duration of 41,251 seconds. We investigate the translation quality and inference speed under two different diffusion sampling schedules: 5 iterations and 10 iterations. The performance metrics, including BLEU scores, Real-Time Factor (RTF), and generation speed (tokens/sec), are summarized in Table 1.

Table 1: Translation benchmark results on the 10K validation set

Target Language	Iterations	BLEU-1	BLEU-2	BLEU-3	BLEU-4	RTF	Tokens/sec
English (EN)	$T = 5$	28.51	18.49	12.31	8.27	0.001	2656.5
	$T = 10$	29.40	19.58	13.48	9.40	0.003	1327.3
Mandarin (ZH)	$T = 5$	27.33	20.98	16.09	12.55	0.001	1705.6
	$T = 10$	29.60	22.80	17.61	13.84	0.003	857.2
Korean (KO)	$T = 5$	9.61	5.49	3.11	1.74	0.001	1719.2
	$T = 10$	10.40	6.12	3.60	2.14	0.003	855.4

As observed, increasing the number of diffusion iterations from $T = 5$ to $T = 10$ yields a consistent improvement in BLEU scores across all language pairs, at the cost of a slightly reduced generation speed. Nevertheless, the system operates significantly faster than real-time in both configurations, validating the extreme efficiency of the non-autoregressive decoding mechanism.

4.3 Hardware Profiling and Edge Deployment Feasibility

To demonstrate the viability of deploying the model on resource-constrained edge devices, we conducted an extensive hardware profiling. The proposed model contains approximately **397.8 million parameters**. Utilizing half-precision floating-point format (FP16/BF16), the theoretical weight footprint is efficiently compressed to just 758.75 MB.

Table 2 details the hardware utilization metrics under different compute setups for a single inference step with batch size = 1 and audio duration about 3.37s.

Table 2: Hardware profiling and performance metrics.

Hardware Setup	**Peak Memory**	**Latency (s)**	**Tokens/sec**	**RTF**
GPU (NVIDIA L4, BF16)	960.00 MB (VRAM)	0.89	14.52	**0.266**
CPU (1 Thread)	4139.30 MB (RAM)	10.68	1.22	3.169
CPU (4 Threads)	4189.43 MB (RAM)	5.50	2.36	1.633

Minimum Edge Requirements: The profiling results indicate that the model is exceptionally lightweight. It necessitates merely $\sim$ **4.53** GB of system RAM (including model weights and runtime buffers) and strictly requires **1.03** GB of VRAM for GPU-accelerated inference. When evaluated on a discrete GPU, the system achieves an RTF of 0.266, confirming its capability to process and translate speech well within real-time constraints. These metrics definitively validate the model's high suitability for on-device applications, strictly satisfying the latency and memory limitations typical of Edge AI hardware (e.g., Qualcomm Snapdragon platforms).

5 Future Work

While our current diffusion speech translation model demonstrates promising capabilities, it is primarily released as a functional prototype due to hardware resource limitations during development. With access to extended computational resources, we plan to implement the following roadmap to significantly enhance the system's performance, efficiency, and versatility:

- **End-to-End Joint Training (Audio and Text):** Currently, the training pipeline only updates the text encoder while the audio encoder remains entirely frozen. In future iterations, we aim to unfreeze the acoustic backbone and implement full end-to-end joint training. This will allow the model to better align complex acoustic features with text embeddings, thereby improving translation accuracy and nuance.

- **Real-Time Streaming Architecture:** To facilitate live communication, we plan to transition from offline, segment-based processing to a continuous streaming methodology. This will involve implementing a **sliding window attention mechanism** over the audio input, allowing the model to generate text incrementally with minimal latency and bounded memory consumption.

- **Multilingual Expansion:** Although the current version supports translating Vietnamese into English, Chinese, and Korean, we intend to scale our training dataset to encompass a significantly wider array of languages. The goal is to build a highly adaptable, multi-directional multilingual speech translation hub.

- **Edge Device Deployment:** Our hardware profiling shows an RTF of 0.266 on a single GPU and a relatively low VRAM requirement ($\sim$1.03 GB). Leveraging this lightweight profile, we plan to package and embed the model for direct deployment on edge devices (e.g., mobile phones, local IoT devices), enabling secure, offline translation without relying on cloud infrastructure.

- **Model Size Optimization and Compression:** To ensure the model runs smoothly on edge hardware with limited CPU/RAM capabilities, we will explore advanced optimization techniques. Methods such as weight quantization (e.g., INT8/INT4), knowledge distillation, and structural pruning will be applied to further reduce the $\sim$397M parameter footprint while preserving the current BLEU score performance.

- **Extension to Speech-to-Speech Translation (Text-to-Speech Integration):** Beyond generating translated text, we plan to extend the pipeline into a direct Speech-to-Speech Translation (S2ST) system. By integrating a diffusion-based Text-to-Speech (TTS) synthesizer, the model will be capable of outputting high-fidelity translated audio directly. Future explorations will also focus on zero-shot voice preservation techniques to maintain the input speaker's vocal identity across multiple target languages while ensuring low-latency inference.

References

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